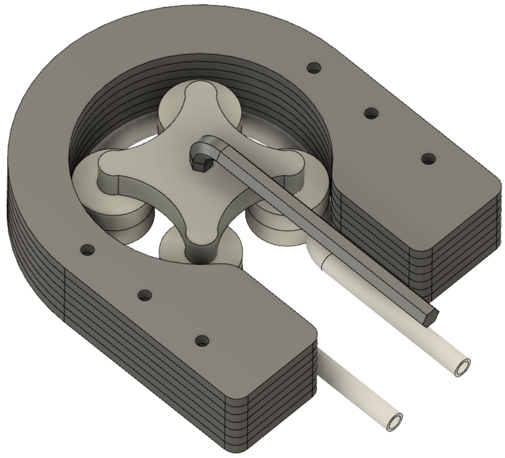

MANUFACTURING INSTRUCTIONS
PERISTALTIC PUMP GROUP 16

TEAM MEMBERS:

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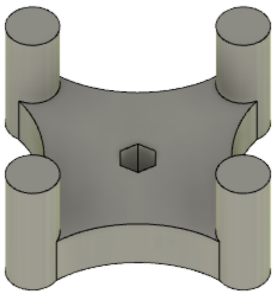
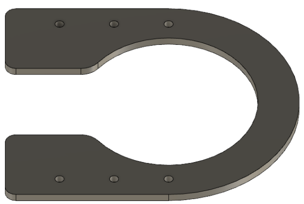
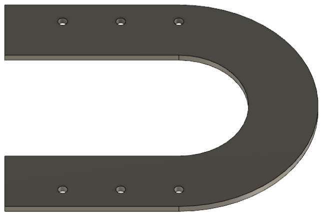
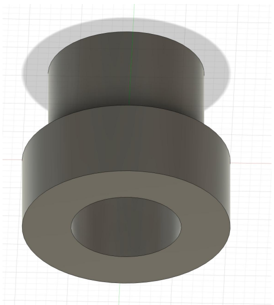
MANUFACTURE LIST

ITEMS	QUANTITY	FILE USED	MANUFACTURING DESCRIPTION
1. Body Rotator 	1	F3d File: spinner STL File: spinner Gcode File:spinner	Material Used: Polyactic Acid (PLA) Machine Used: Ultimaker 3D Printer 1. Keep the same default settings preset on the 3D printer. 2. Transfer Gcode file to the Ultimaker 3D Printer print the body rotator. 3. Alternatively, (manual ways): export the STL File to the Ultimaker Cura Software, and transfer it to the Gcode file. 4. Apply glue on the 3D Printer Plate before the process starts and start the printing. Wait until the printing is done and allow the body rotator to cool down for 20 minutes before taking it from the printer. 5. When printing, ensure the position of the body rotator as per *diagram attached to the items column.*
2. Casing 	7	F3d File: Base .dxf File: Base	Material Used: 4mm Arcylic Sheet Machine Used: 3D Printing 1. Export the design from Fusion 360 to .dxf format. 2. Optimally arrange the sheets and print location to reduce the waste used. 3. Start the machine and apply safety measures while handling the machine. 4. Produce 7 pcs of casings, which will be stacked together as a whole casing with 28mm of height.

3. Case Cover 	1	F3d File: Cover .dxf File: Cover	Material Used: 4mm Arcylic Sheet Machine Used: 3D Printing 1. Export the design from Fusion 360 to .dxf format. 2. Optimally arrange the sheets and print location to reduce the waste used. 3. Start the machine and apply safety measures while handling the machine. 4. Produce 1 pcs of case cover, which will be stacked on top of the covers and bonded with the screw and bolts.
4. Cap 	2	.F3d File: The cap STL File:The cap Gcode File: UM3_The cap	Material Used: Polyactic Acid (PLA) Machine Used: Ultimaker 3D Printer 1. Keep the same default settings preset on the 3D printer. 2. Transfer Gcode file to the Ultimaker 3D Printer to print the caps. 3. Alternatively, (manual ways): export the STL File to the Ultimaker Cura Software, and transfer it to the Gcode file. 4. Apply glue on the 3D Printer Plate before the process starts and start the printing. Wait until the printing is done and allow the caps to cool down for 20 minutes before taking it from the printer. 5. When printing, ensure the position of the caps per *diagram attached to the items column.* 6. Produce 2 pcs of the caps, which will be used in connecting the bags mouth to the tube, to ensure no leakings when fluid flows.

BILLS OF MATERIALS

No	Item	Units	Link	Cost
1	Allen key	1	DT000215 Duratool, Hex Key, 6 mm, Long Arm | Farnell UK	£0.95
2	Bearings	8	uk.rs-online.com/basket?source=web	£11.20
3	RS PRO Pozidriv Countersunk A2 304 Stainless Steel Machine Screws DIN 965, M4x40mm	6	RS PRO Pozidriv Countersunk A2 304 Stainless Steel Machine Screws DIN 965, M4x40mm | RS (rs-online.com)	£0.90
4	Saint Gobain Versilic® Flexible Tube, Silicone, 5mm ID, 7mm OD, Clear, 1m	1	760410 | Saint Gobain Versilic® Flexible Tube, Silicone, 5mm ID, 7mm OD, Clear, 25m | RS (rs-online.com)	£2.10
5	Bags	2	Container Lotion Dispenser Bag Liquid Dispensing Bag Refillable Pouches | eBay	£6.72
6	Breadboard	1	Provided	£0.00
7	Screwdriver	1	Provided	£0.00

Safety instructions:

1. Allow the final cut out to cool down before taking off.
2. Ensure that the laser cutting machine is properly vented to remove any toxic gases
3. Only four students and one technician are allowed in the laser cutting room.
4. Apply all the safety measures when handling the 3D Printer, Laser Cutter and when doing the assembly of the product.